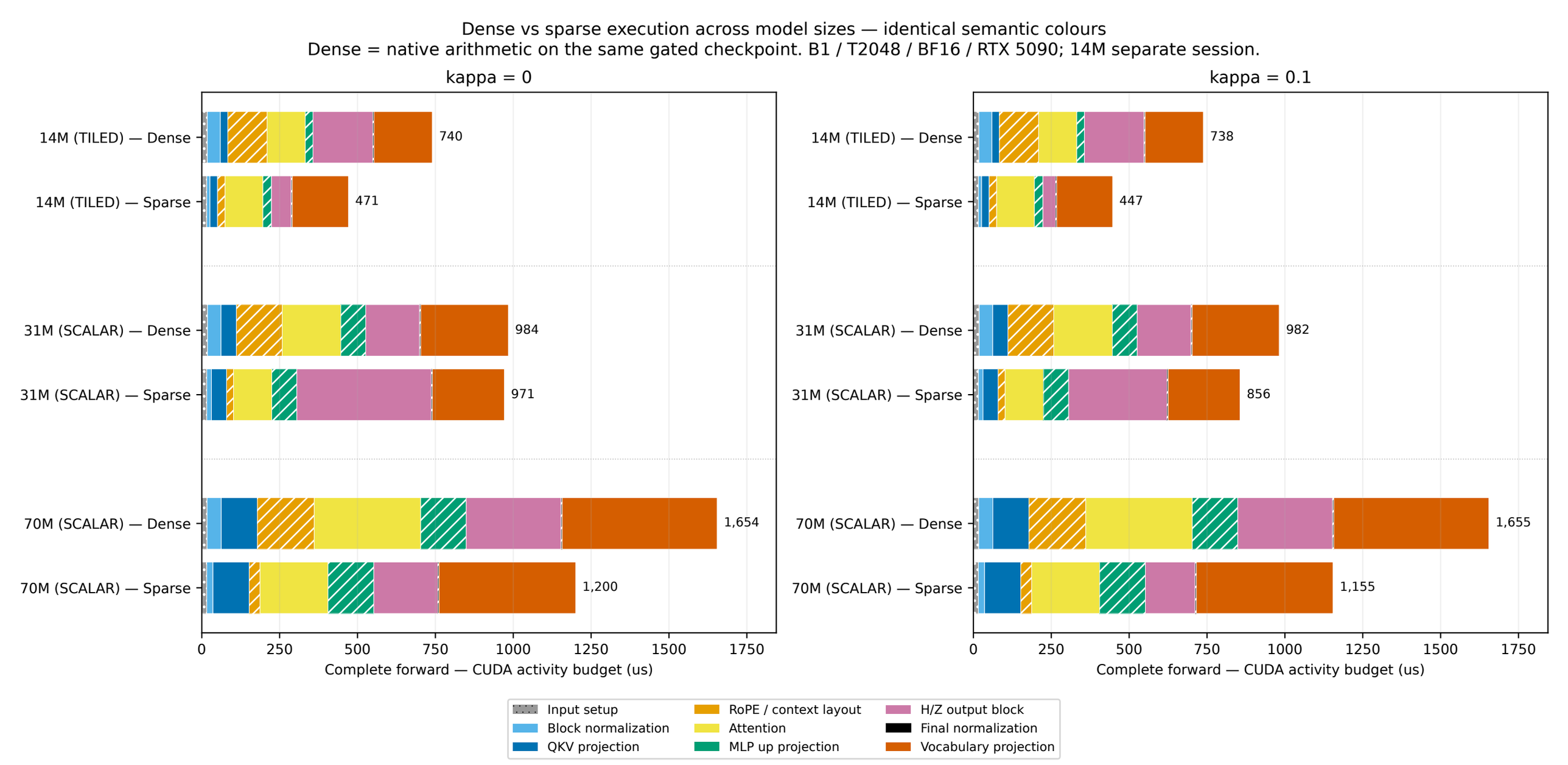

Dense vs sparse execution across model sizes — identical semantic colours
Dense = native arithmetic on the same gated checkpoint. B1 / T2048 / BF16 / RTX 5090; 14M separate session.
kappa = 0
kappa = 0.1
14M (TILED) — Dense
14M (TILED) — Sparse
31M (SCALAR) — Dense
31M (SCALAR) — Sparse
70M (SCALAR) — Dense
70M (SCALAR) — Sparse
740
471
984
971
1,654
1,200
738
447
982
856
1,655
1,155
0
250
500
750
1000
1250
1500
1750
Complete forward — CUDA activity budget (us)
Input setup
Block normalization
QKV projection
RoPE / context layout
Attention
MLP up projection
H/Z output block
Final normalization
Vocabulary projection